Gradient Descent Method

STAT 525 - Topics in Computational Statistics

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Unconstrained Optimization

- In the first week, we show that most statistical and machine learning models can be cast as an optimization problem

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & x \in \Omega \end{aligned}$$

- In this section, we focus on a special case
  - Here, $\Omega = \mathbb{R}^d$. That is why we called it unconstrained problem
  - $f(x)$ is convex and twice differentiable

Unconstrained Optimization

- Because $f(x)$ is convex and differentiable, a necessary and sufficient condition for a point x^* to be optimal is

$$\nabla f(x^*) = 0$$

- In other word, we are going to look for some point x^* that has small gradient

Descent Methods

- In order to look for the optimal point x^*, people usually adopt iterative methods. Specifically, we start with a point $x^{(0)}$ and keep updating the point's location

$$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}, \qquad k = 0, 1, \ldots$$

- Here, $t^{(k)} > 0$ is step size (aka learning rate) and $\Delta x^{(k)}$ is search direction.
- An method is called s descent method if

$$f(x^{(k+1)}) < f(x^{(k)}).$$

This condition requires

$$\nabla f(x^{(k)})^T \Delta x^{(k)} < 0.$$

Descent Methods

The general procedure of descent method can be summarized as

1. Determine a search direction $\Delta x^{(k)}$
2. Find a step size $t^{(k)} > 0$
3. Update the point $x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$
4. Check the stopping criteria, e.g. $\|\nabla f(x^{(k+1)})\|$ is small enough

- Different descent methods mean different ways to choose the direction or different ways to choose step size
- Δx is a descent direction for f at x if $f(x + t\Delta x) < f(x)$ for all $t > 0$ sufficiently small.

Gradient Descent

- For the direction Δx, we have

$$\lim_{t \to 0} \frac{f(x + t\Delta x) - f(x)}{t} = \nabla f(x)^T \Delta x$$

- A reasonable choice to reduce the objective function is

$$\Delta x = -\nabla f(x)$$

  as this direction reduces rate of objective value in the largest magnitude.
- With this search direction, we can introduce gradient descent method (GD) (a.k.a. steepest descent)

$$x^{(k+1)} = x^{(k)} - t^{(k)} \nabla f(x^{(k)}).$$

Smooth Function

- After deciding the direction, how shall we choose the step size and what is the convergence rate?
- A commonly used assumption is f is M-smooth. By descent lemma,

$$f(x^{(k+1)}) \leq f(x^{(k)}) - t^{(k)} \nabla f(x^{(k)})^T \nabla f(x^{(k)}) + {t^{(k)}}^2 \frac{M}{2} \|\nabla f(x^{(k)})\|^2$$

Minimizing the side of right hand yields $t^{(k)} = 1/M$ and

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2M} \|\nabla f(x^{(k)})\|^2. \tag{1}$$

This is the fundamental inequality in analysis of optimization problem.

Smooth Function

- If f is M-smooth, $f(x) \geq \bar{f}$ for any x, and $t^{(k)} = 1/M$, then

$$\min_{0 \leq i \leq k-1} \|\nabla f(x^{(i)})\| \leq \sqrt{\frac{2M(f(x^{(0)}) - \bar{f})}{k}}$$

- The convergence is slow.

Proof: Applying (1) repeatedly yields

$$f(x^{(k)}) \leq f(x^{(0)}) - \frac{1}{2M} \sum_{i=0}^{k-1} \|\nabla f(x^{(i)})\|^2.$$

Because $f(x^{(k)}) \geq \bar{f}$, we have

$$\sum_{i=0}^{k-1} \|\nabla f(x^{(i)})\|^2 \leq 2M(f(x^{(0)}) - \bar{f}).$$

This immediately suggests

$$\min_{0 \leq i \leq k-1} \|\nabla f(x^{(i)})\| \leq \sqrt{\frac{1}{k} \sum_{i=0}^{k-1} \|\nabla f(x^{(i)})\|^2} \leq \sqrt{\frac{2M(f(x^{(0)}) - \bar{f})}{k}}.$$

Convex and Smooth Function

- In this convex and smooth case, how fast can we find the optimal solution? If f is M-smooth and we still choose step size $1/M$, then

$$\|x^{(k+1)} - x^*\|^2 \leq \|x^{(k)} - x^*\|^2 - \frac{1}{M^2}\|\nabla f(x^{(k)})\|^2.$$

Convex and Smooth Function

Proof: Since $x^{(k+1)} = x^{(k)} - \nabla f(x^{(k)})/M$ and f is smooth, we have

$$
\begin{aligned}
&\|x^{(k+1)} - x^*\|^2 \\
=&\|x^{(k)} - x^* - (\nabla f(x^{(k)}) - \nabla f(x^*))/M\|^2 \\
=&\|x^{(k)} - x^*\|^2 - \frac{2}{M}\langle x^{(k)} - x^*, \nabla f(x^{(k)}) - \nabla f(x^*)\rangle + \frac{1}{M^2}\|\nabla f(x^{(k)})\|^2 \\
\leq&\|x^{(k)} - x^*\|^2 - \frac{2}{M^2}\|\nabla f(x^{(k)})\|^2 + \frac{1}{M^2}\|\nabla f(x^{(k)})\|^2 \\
=&\|x^{(k)} - x^*\|^2 - \frac{1}{M^2}\|\nabla f(x^{(k)})\|^2
\end{aligned}
$$

Convex and Smooth Function

- If f is smooth and choose step size $1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \frac{2M\|x^{(0)} - x^*\|^2}{k}.$$

- It is faster than smooth only case.

Proof: We already have

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2M}\|\nabla f(x^{(k)})\|^2.$$

Since $f(x^{(k)}) - f(x^*) \leq \nabla f(x^{(k)})^T(x^{(k)} - x^*) \leq \|\nabla f(x^{(k)})\|\|x^{(k)} - x^*\|$, we have

$$f(x^{(k+1)}) - f(x^{(k)}) \leq -\frac{1}{2M}\frac{(f(x^{(k)}) - f(x^*))^2}{\|x^{(k)} - x^*\|^2} \leq -\frac{1}{2M}\frac{(f(x^{(k)}) - f(x^*))^2}{\|x^{(0)} - x^*\|^2}$$

This leads to

$$\frac{1}{f(x^{(k+1)}) - f(x^*)} - \frac{1}{f(x^{(k)}) - f(x^*)} \geq \frac{1}{2M\|x^{(0)} - x^*\|^2}\frac{f(x^{(k)}) - f(x^*)}{f(x^{(k+1)}) - f(x^*)}.$$

Thus,

$$\frac{1}{f(x^{(k)}) - f(x^*)} \geq \frac{k}{2M\|x^{(0)} - x^*\|^2}.$$

Strongly Convex and Smooth Function

- We generalize the convex function into strongly convex and smooth function.
- With strongly convexity, we have the "sandwiching" effect

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{M}{2}\|y - x\|^2$$

and

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{m}{2}\|y - x\|^2.$$

Strongly Convex and Smooth Function

- In strongly convex function, the norm of gradient provides useful information about how far away we are from optimality.
- Minimizing both sides in strongly convex condition yields

$$f(x^*) \geq f(x) - \frac{1}{2m}\|\nabla f(x)\|^2.$$

- In other words,

$$\|\nabla f(x)\|^2 \geq 2m(f(x) - f(x^*)).$$

Strongly Convex and Smooth Function

- Similarly, we have

$$\begin{aligned} f(x^*) &\geq f(x) + \langle \nabla f(x), x^* - x \rangle + \frac{m}{2}\|x^* - x\|^2 \\ &\geq f(x) - \|\nabla f(x)\|\|x^* - x\| + \frac{m}{2}\|x^* - x\|^2. \end{aligned}$$

- So

$$\|x^* - x\| \leq \frac{2}{m}\|\nabla f(x)\|.$$

Strongly Convex and Smooth Function

- If f is a strongly convex and smooth function and we choose the step size $t^{(k)} = 1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{m}{M}\right)^k \left(f(x^{(0)}) - f(x^*)\right),$$

where x^* is the optimal solution.

Strongly Convex and Smooth Function

Proof: First, we note that

$$f(x^{(k+1)}) \leq f(x^{(k)}) - \frac{1}{2M}\|\nabla f(x^{(k)})\|^2 \leq f(x^{(k)}) - \frac{m}{M}(f(x^{(k)}) - f(x^*)).$$

Rearranging the terms to have

$$f(x^{(k+1)}) - f(x^*) \leq \left(1 - \frac{m}{M}\right)(f(x^{(k)}) - f(x^*)).$$

Relax Strongly Convexity

- Strong convexity can lead to linear convergence, but it might not hold in practice.
- Consider an example: logistic regression. We observe (X_i, Y_i), $i = 1, \ldots, n$, where $Y_i \in \{-1, 1\}$ (equivalently we can also write $Y_i \in \{0, 1\}$)

$$\mathbb{P}(Y_i = 1) = 1 - \mathbb{P}(Y_i = -1) = \frac{1}{1 + \exp\left(-\beta^T X_i\right)}.$$

- The maximum likelihood estimate can then be formulated as the following optimization problem

$$\min_{\beta} f(\beta) = \frac{1}{n} \sum_{i=1}^{n} \log\left(1 + \exp(-Y_i \beta^T X_i)\right).$$

Relax Strongly Convexity

- Since $f(\beta)$ is twice-differentiable, the Hessian matrix can be wrote as

$$\nabla^2 f(\beta) = \frac{1}{n} \sum_{i=1}^{n} \frac{\exp(-Y_i \beta^T X_i)}{(1 + \exp(-Y_i \beta^T X_i))^2} X_i X_i^T$$

  Clearly when $\beta \to \infty$, then

$$\nabla^2 f(\beta) \to 0.$$

- We discuss how condition of strong convexity can be relaxed
  - Local strong convexity
  - Regularity condition
  - Polyak-Lojasiewicz condition

Local Strong Convexity

- A twice-differentiable function f is said to be local strongly convex and smooth if there exists m and M such that

$$mI \preceq \nabla^2 f(x) \preceq MI, \qquad x \in B_0.$$

  Here, $B_0 = \{x : \|x - x^*\| \leq \|x^{(0)} - x^*\|\}$.
- If we replace strongly convex and smooth condition by local strongly convex and smooth condition, all the conclusions mentioned before still hold.
- The idea is that we always have $\|x^{(t+1)} - x^*\| < \|x^{(t)} - x^*\|$, so $x^{(t)}$ won't escape from B_0.

Local Strong Convexity

- In the example of logistic regression, we can expect there exist m and M such that

$$mI \preceq \nabla^2 f(\beta) \preceq MI, \qquad \|\beta - \beta^*\| \leq \|\beta^{(0)} - \beta^*\|.$$

- We can still achieve linear convergence if we apply GD to find MLE of logistic regression.

Regularity Condition

- The m-strongly convex and M-smooth condition can be written in an equivalent way

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq m \|x - y\|^2$$

and

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \frac{1}{M} \|\nabla f(x) - \nabla f(y)\|^2$$

for any pair of x and y.

- In the proof, the conditions can be relaxed as, for any x,

$$\langle \nabla f(x) - \nabla f(x^*), x - x^* \rangle \geq \frac{m}{2} \|x - x^*\|^2 + \frac{1}{2M} \|\nabla f(x) - \nabla f(x^*)\|^2$$

which is equivalent to the following regularity condition

$$\langle \nabla f(x), x - x^* \rangle \geq \frac{m}{2} \|x - x^*\|^2 + \frac{1}{2M} \|\nabla f(x)\|^2$$

Regularity Condition

- If we replace strongly convex and smooth condition by regularity condition, all the conclusions mentioned before still hold.

Polyak-Lojasiewicz Condition

- We introduce another condition to replace strongly convex condition, Polyak-Lojasiewicz (PL) condition

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq m\left(f(x) - f(x^*)\right).$$

- It means the gradient grows faster than a quadratic function as x moves away from x^*.
- Suppose f satisfies PL condition and is smooth. If we choose the step size as $t^{(k)} = 1/M$, then

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{m}{M}\right)^k \left(f(x^{(0)}) - f(x^*)\right)$$

Polyak-Lojasiewicz Condition

Proof: Since $x^{(k+1)} = x^{(k)} - \nabla f(x^{(k)})/M$ and f is M-smooth, we have

$$\begin{aligned} f(x^{(k+1)}) &\leq f(x^{(k)}) - \langle \nabla f(x^{(k)}), \nabla f(x^{(k)})/M \rangle + \frac{M}{2} \|\nabla f(x^{(k)})/M\|^2 \\ &\leq f(x^{(k)}) - \frac{1}{2M} \|\nabla f(x^{(k)})\|^2 \end{aligned}$$

By using PL condition, we have

$$f(x^{(k+1)}) - f(x^*) \leq f(x^{(k)}) - f(x^*) - \frac{m}{M} \left(f(x^{(k)}) - f(x^*) \right)$$

This suggests that

$$f(x^{(k)}) - f(x^*) \leq \left(1 - \frac{m}{M}\right)^k \left(f(x^{(0)}) - f(x^*) \right)$$

Polyak-Lojasiewicz Condition

$$\frac{1}{2}\|\nabla f(x)\|^2 \geq m\left(f(x) - f(x^*)\right).$$

- PL condition doesn't suggest f is convex or strongly convex
- PL condition doesn't suggest x^* is unique, but suggests that every stationary point is a global minimum

Polyak-Lojasiewicz Condition

- Suppose we have data (X_i, Y_i), $i = 1, \ldots, n$. Here, $X_i \in \mathbb{R}^d$. In least square estimation of linear model, we have

$$\min_{\beta} f(\beta) = \frac{1}{2} \sum_{i=1}^{n} (Y_i - X_i^T \beta)^2.$$

- The Hessian matrix of f is

$$\nabla^2 f(\beta) = \sum_{i=1}^{n} X_i X_i^T$$

- When $d > n$ (high-dimension case), f is convex but not strongly convex as $\nabla^2 f(\beta)$ is rank n.

Polyak-Lojasiewicz Condition

- However, we can show than PL condition can hold when $d > n$ and thus linear convergence is achievable.
- Since $\nabla f(\beta) = \sum_{i=1}^{n}(Y_i - X_i^T\beta)X_i$,

$$\begin{aligned}
\|\nabla f(\beta)\|^2 &= \left(\sum_{i=1}^{n}(Y_i - X_i^T\beta)X_i\right)^T \left(\sum_{i=1}^{n}(Y_i - X_i^T\beta)X_i\right) \\
&= \sum_{i,j=1}^{n}(Y_i - X_i^T\beta)X_i^T X_j(Y_j - X_j^T\beta) \\
&\geq \lambda_{\min}(A)\sum_{i=1}^{n}(Y_i - X_i^T\beta)^2 \\
&= 2\lambda_{\min}(A)\left(f(\beta) - f(\beta^*)\right)
\end{aligned}$$

where A is gram matrix of X_i, i.e. $A_{i,j} = X_i^T X_j$. We use the fact $f(\beta^*) = 0$ as $d > n$.

General Descent Method

The general procedure of descent method can be summarized as

1. Determine a search direction $\Delta x^{(k)}$
2. Find a step size $t^{(k)} > 0$
3. Update the point $x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}$
4. Check the stopping criteria, e.g. $\|\nabla f(x^{(k+1)})\|$ is small enough

General Descent Method: Search Direction

- The direction is not necessary to be negative gradient, as sometimes there are more efficient directions and gradient could be unavailable. But we still hope the search direction is somehow similar to the negative gradient, i.e.,

$$\frac{-(\Delta x^{(k)})^T \nabla f(x^{(k)})}{\|\Delta x^{(k)}\| \|\nabla f(x^{(k)})\|} \geq \bar{\epsilon} > 0$$

and

$$\gamma_1 \leq \frac{\|\Delta x^{(k)}\|}{\|\nabla f(x^{(k)})\|} \leq \gamma_2.$$

- Here, we provide several possible options.

General Descent Method: Search Direction

- The transformed negative gradient direction

$$\Delta x^{(k)} = -S^{(k)} \nabla f(x^{(k)})$$

  where $S^{(k)}$ is a symmetric positive definite matrix with eigenvalues in $[\gamma_1, \gamma_2]$.
- Newton's method ($S^{(k)} = \nabla^2 f(x^{(k)})^{-1}$) and Quasi-Newton method can be seen as special cases.

General Descent Method: Search Direction

- The Gauss–Southwell variant of coordinate descent

$$\Delta x^{(k)} = -[\nabla f(x^{(k)})]_{i^{(k)}} e_{i^{(k)}},$$

  where $i^{(k)} = \text{argmax}_i |[\nabla f(x^{(k)})]_i|$.
- Randomized coordinate descent: we pick up the coordinate $i^{(k)}$ randomly.
- We are going to talk about coordinate descent methods in details later.

General Descent Method: Search Direction

- Stochastic gradient descent methods

$$\Delta x^{(k)} = -g(x^{(k)}, \xi^{(k)}),$$

where

$$\mathbb{E}\left(g(x^{(k)}, \xi^{(k)})\right) = \nabla f(x^{(k)}).$$

General Descent Method: Step Size

- Given direction Δx, how shall we choose step size? We introduce four options
  - Fixed step size: very popular in practice and performs good
  - Exact line search
  - Approximate line search
  - Backtracking line search

General Descent Method: Exact Line Search

- Exact line search

$$t^{(k)} = \underset{s \geq 0}{\operatorname{argmin}} f(x^{(k)} + s\Delta x^{(k)}).$$

Basically, we are looking for the optimal points on a ray $\{x^{(k)} + s\Delta x^{(k)} | s \geq 0\}$. This strategy is usually used when the minimization problem on a ray is computationally efficient.

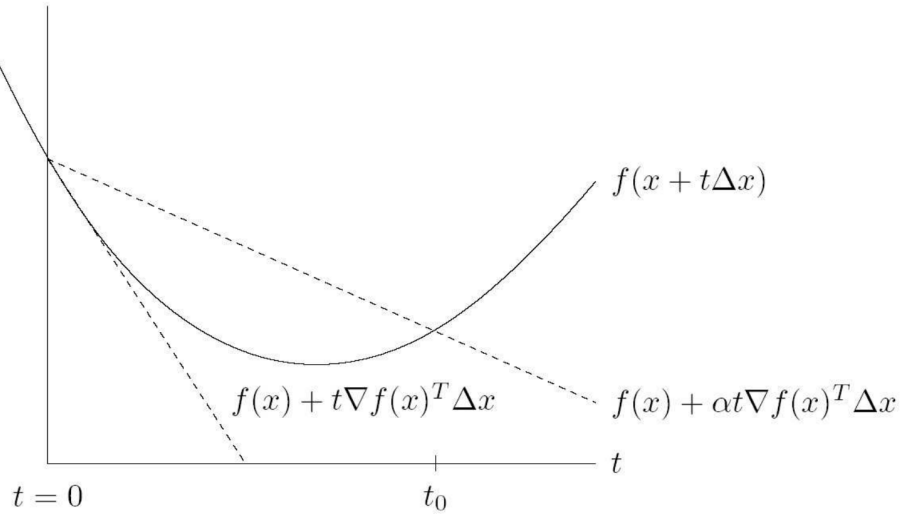
General Descent Method: Approximate Line Search

- Finding the optimal point in exact line search is usually not necessary, as it is generally not an easy job. Most of time, it is sufficient to find an approximately optimal point or a point that reduce f enough.
- A simple and effective scheme: approximate line search
- We first introduce weak Wolfe conditions: for $0 < c_1 < c_2 < 1$,

$$f(x^{(k)} + t\Delta x^{(k)}) < f(x^{(k)}) + c_1 t \nabla f(x^{(k)})^T \Delta x^{(k)}$$
$$\nabla f(x^{(k)} + t\Delta x^{(k)})^T \Delta x^{(k)} > c_2 \nabla f(x^{(k)})^T \Delta x^{(k)}.$$

First condition is to ensure the step is not too long and the second to ensure the step is not too short.

General Descent Method: Approximate Line Search



- The first condition ensure sufficient decrease

General Descent Method: Approximate Line Search

- Extrapolation-Bisection Line Search
  1. Initialize $L = 0$, $U = +\infty$, and $t = 1$
  2. Check first condition. If

$$f(x^{(k)} + t\Delta x^{(k)}) > f(x^{(k)}) + c_1 t \nabla f(x^{(k)})^T \Delta x^{(k)},$$

     then set $U = t$ and $t = (U + L)/2$. Return step 2.
  3. If the first condition is satisfied, check the second condition. If

$$\nabla f(x^{(k)} + t\Delta x^{(k)})^T \Delta x^{(k)} < c_2 \nabla f(x^{(k)})^T \Delta x^{(k)},$$

     then set $L = t$ and $t = 2L$ $(U = +\infty)$ or $(U + L)/2$ $(U < +\infty)$. Return step 2.
  4. If both conditions are satisfied, stops.

General Descent Method: Backtracking Line Search

- A more simple and effective scheme: only check the first condition

$$f(x^{(k)} + t\Delta x^{(k)}) < f(x^{(k)}) + c_1 t \nabla f(x^{(k)})^T \Delta x^{(k)}.$$

  We don't worry the step is too short as we start with a large step.
- Backtracking line search
  1. Initialize $t = 1$ and $0 < \beta < 1$
  2. Check stopping condition.
  3. If stopping condition is not satisfied, then $t = \beta t$ and return step 2.